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Effective use of fibres of various types and material for shotcrete in rock support for tunnels

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ABSTRACT: Large volumes of steel fibre reinforced shotcrete (sprayed concrete) and steel bolts are commonly used to support tunnels in hard rock. This generates a high CO_2-footprint which must be reduced in order to decelerate the continuously increasing average temperature worldwide caused by the emissions of greenhouse gases. Thus, alternative design methods and the possibility to use other materials than steel are currently investigated. Work is ongoing on testing the load-bearing capacity of shotcrete reinforced with fibres of steel, basalt and synthetic materials. This also includes a comparison between tests using Round Determinate Panels (RDP) and four-point bending of beams. Moreover, the practical use of RDP testing as a quality control methodology is also investigated in situ. Here, the goal is to identify several shotcrete mixes suitable for use in tunnelling so that the right material and fibre volume can be used in the right place.

1 BACKGROUND

The world is facing an enormous challenge in which the emission of greenhouse gases (GHG) must be reduced in order to decelerate the continuously increasing average temperature worldwide. At the same time, a long trend of urbanisation have led to that the infrastructure network in many densely populated cities must be expanded in order to have a functional transportation infrastructure. Thus, the road and railway network must be expanded which includes bridges and tunnels. In many cases, tunnels are the most favourable option since they do not interfere with the existing cityscape. Tunnels in soft ground are often supported by cast reinforced concrete, while tunnels in hard rock normally are supported with a combination of fibre-reinforced shotcrete, i.e. sprayed concrete, in combination with rock bolts.

In Sweden, most tunnels are constructed in hard and jointed rock of good quality and hence, fibre-reinforced shotcrete and rock bolts is commonly used as rock support. Moreover, road and railway tunnels are owned by the Swedish Road Administration (SRA) and while the design of tunnels is based on Eurocode, supplementary rules are issued by the SRA. Furthermore, the SRA also decides the type of materials, e.g. type of reinforcement and rock bolts, that are allowed in the tunnels. As for today, only steel fibres and steel rock bolts are allowed. One reason for this is the long experience with research and practical use of steel fibres in tunnels that started in the 1980's by the work of (Holmgren, 1979) and have continued with work presented by e.g. (Ansell, 1999), (Malmgren, 2005), (Nordström, 2005) and (Sjölander, 2020).

Today, the market offers many alternative materials to steel, such as polypropylene, glass fibre and basalt fibre, which do not suffer from corrosion and have the potential of lowering the emissions of GHG during construction. Therefore, the SRA have issued two research projects in order to investigate the possibilities of reducing the environmental impact from the construction and life-cycle of the tunnels in Sweden. In the first, the feasibility of using glass fibre bolts

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as permanent rock bolts was investigated by (Johansson et al., 2020). The second project is currently on-going and focuses on the advantages and disadvantages of steel, synthetic and basalt fibres. An overview of the topics the research project aims to cover is shown in Figure 1.

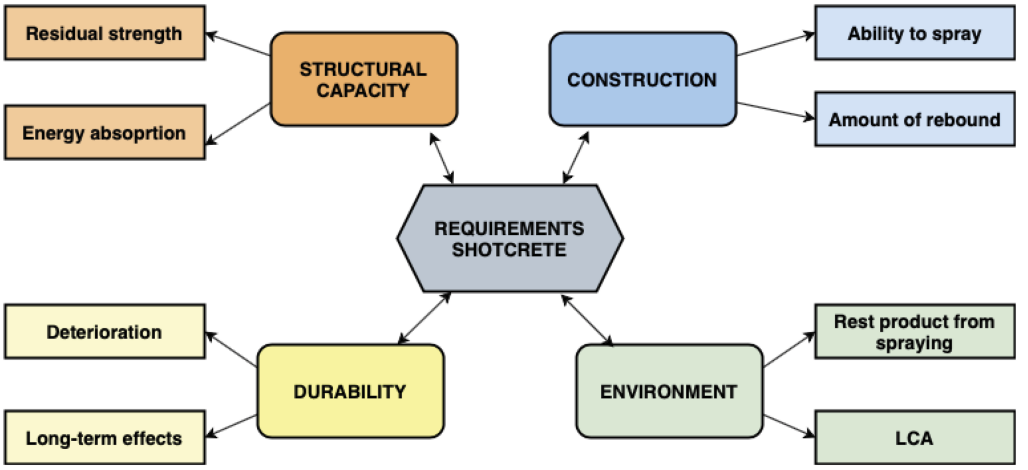


Figure 1. Overview of the research focus in the project.

Obviously, the structural capacity is an important topic but also the design philosophy, which stipulates the structural requirements put on the shotcrete, normally in terms of an required energy absorption or a minimum residual flexural strength. On the practical side, the ability to spray the required dosages of the different fibres, and if large variations in rebound exist between different fibre types is also of interest. To assess the fibres impact on the environment, tools such as life-cycle assessment and environmental product declaration (EPD) should be used. However, it is also important to consider how the mock, i.e. fibre-reinforced shotcrete that rebound from the tunnel surface, should be taken care of. In Norway, large amounts of synthetic fibres ended up floating in the fjords, which lead to a complete ban of this fibre type for all Norwegian road and railway tunnels (Myren, 2018). Finally, the knowledge regrading how different fibres react under long-term loading and their durability must be investigated. The latter is the most challenging aspect of the project. It includes investigating which possible deterioration mechanisms exist for the different fibres, and if they may be initiated or accelerated in a tunnel environment.

Within this project, the structural capacity of different fibre types have been investigated by experimental testing and reported by (Rengarajan, 2020) and (Sjölander et al., 2022a). The environmental impact of different fibre types was investigated in a case study reported by (Anand, 2021), while (Brodd and Östlund, 2022) investigate the impact of different test methods for the structural capacity of shotcrete and the possibilities of replacing part of the cement with slagg. In this paper, results from the experimental campaign which contain structural testing of fibre-reinforced shotcrete with various fibre types and dosages performed on cast and sprayed beams according to EN 14488-3 (CEN, 2006a) and round determinate panels according to ASTM C1550 (ASTM, 2019) are presented together with the ongoing work.

2 DESIGN OF SHOTCRETE ROCK SUPPORT

The design of tunnels and other geotechnical structures is complex due to the many uncertainties that surrounds the quality of naturally occurring building materials such as rock. Therefore, the design rules in Europe, i.e. Eurocode 7 (CEN, 2004), gives the designer a flexibility to verify the structural capacity using different methodologies, i.e. the observational method, numerical methods, empirical methods as well as experimental test and models. For tunnels in

hard rock, the rock mass is part of the support system due to composite action between shotcrete, rock and rock bolts but will also act as the load on the support. The total load on the support system is difficult to predict since it depends on the time of installation as well as the load distribution between shotcrete and rock. Therefore, the design approach and methodology differs between different countries but also between different designers.

2.1 *Post-cracking performances of fibre reinforced shotcrete*

For fibre-reinforced shotcrete, the post-cracking performance, i.e. residual strength or energy absorption, is governed by the energy consumed while fibres are being deformed and pulled-out from the shotcrete (Nordström, 2005; Sjölander, 2020). Thus, the geometry and mechanical properties of the fibre together with the bond strength between the fibre and shotcrete matrix will all effect the post-cracking performance. Different type of fibres will therefore be efficient in different ways. In general, steel and basalt fibres are more efficient at lower deflections, while synthetic fibres are more efficient at larger deformations. However, this is a generalised explanation and the structural behaviour also depends, as mentioned above, on the length and profile of the fibres which could be altered in the manufacturing process. Due to this, different fibre materials may be more or less efficient depending on which structural requirements the shotcrete must fulfil. Structural requirements are either based on the residual flexural strength or the energy absorption capacity and is determined based on which design approach is used.

2.2 *Empirical and numerical design*

The most widely used empirical design methods for rock support is the Q-method developed by Barton et al. (Barton, 1974), and the Rock Mass Rating (RMR) system developed by (Bieniawski, 1973). In both methods, the quality of the rock mass is estimated through a geotechnical inspection of the excavated rock mass. Based on a fixed number of parameters, a numerical value which represent different qualities of the rock mass is calculated. Based on this, both methods suggests a rock support, i.e. a required shotcrete thickness and spacing between rock bolts. The Q-system also gives a minimum energy absorption based on testing of square panels according to EN-14488-5 (CEN, 2006b). Besides the recommendations by the Q-system, national recommendations with respect to energy absorption or residual flexural strength are sometimes used, as summarised by (Sjölander et al., 2022a).

If the design should be verified by numerical calculations, local failure modes in which a loose block is loading the shotcrete is commonly assumed, see e.g. (Barrett and McCreath, 1995). Based on this, the required thickness and residual flexural strength of the fibre-reinforced shotcrete (FRS) can be determined. Here, the performance of the shotcrete is normally tested on concrete beams, e.g. EN 14488-3 (CEN, 2006a), and the residual strength is determined for a vertical displacement in the range between one and four millimeters. It is known that a large scatter in the residual strength exist from tests based on beams due to the small fracture surface. Moreover, (Brodd and Östlund, 2022) performed numerical simulations to show that the dosage of fibres could be reduced significantly if a trustworthy correlation between energy absorption measured on round determinate panels and residual flexural strength measured on beams could be established.

3 EXPERIMENTAL WORK

Experimental results for fibre-reinforced shotcrete with steel and synthetic fibres are widely available in the literature, see e.g. (Barros et al., 2005) and (Bernard, 2002). Results with basalt fibres are less common but have recently gained an increased interest from the scientific community, see e.g. (Mohaghegh, 2018). However, most results contain only one type of fibre and in other cases, the shotcrete mix was not similar to the mix that is traditionally used in Swedish tunnels. Therefore, a test series with different type of fibres and a traditionally Swedish shotcrete mix was performed to facilitate the comparison between different fibres. Another

important reason for conducting this test series was to investigate if any difference in structural behaviour or fibre orientation could be observed between cast and sprayed samples. The results for the fibre orientation is presented by (Ansell, 2021; 2023). Below, the test set-up and the evaluation of the structural and environmental performance is presented.

3.1 *Test set-up*

To investigate how different type of fibres affect the structural capacity of fibre-reinforced shotcrete, an experimental campaign was conducted at the laboratory of Vattenfall R&D in Älvkarleby, Sweden. Tests were performed with two different types of steel fibres, one synthetic fibre and one basalt fibre type. Both cast and sprayed specimens have been tested and the complete test matrix is presented in Table 1. The complete results from the test series are presented by (Sjölander et al., 2022b).

Table 1. Test matrix for the experimental campaign.

			Cast			Sprayed		
Fibre	Material	Dosage kg/m^3	Cubes [-]	Beams [-]	Panels [-]	Cubes [-]	Beams [-]	Panels [-]
Dramix 3D	Steel	30/40/50	3/3/3	3/3/3	3/3/3	3/3/3	-/-/-	3/3/3
Dramix 4D	Steel	30/40/50	3/3/3	3/3/3	3/3/3	3/3/3	-/-/-	3/3/3
Basalt Minibar	Basalt	14/16/20	3/3/3	3/3/3	3/3/3	3/3/3	-/-/-	3/3/3
BarChip 54	Synthetic	3/6/9	3/3/3	3/3/3	3/3/3	3/3/3	-/-/-	3/3/3



Figure 2. From left to right, casting, storage of specimens during the first 24 hours, and storage until testing.

Table 2. Shotcrete mix used in the experiments.

Material	Content	Unit	Comment
Cement	500	kg/m^3	
Water	204	kg/m^3	
w/c	0.41	-	
Aggregate 0-2 mm	489	kg/m^3	
Aggregate 0-8 mm	1141	kg/m^3	
Release agent	1.6	kg/m^3	Sika Perfin 301
Plasticizer	5.0	kg/m^3	Sika Viso-Create 6730
Air	4.5-5.1	%	Given as range for all samples
Slump	80-150	mm	Given as range for all samples

The shotcrete mix is presented in Table 2 and represent a typical mixture used in Swedish tunnels. The compressive strength was tested on cubes according to EN 12390-3 (CEN, 2002), the residual flexural strength tests were based on EN 14488-3 (CEN, 2006a) and the energy absorption was evaluated using round determinate panels (RDP), based on ASTM C1550 (ASTM, 2019). For each dosage of fibres, all test speciemens were casted or sprayed from one batch. Cast specimens were produced by the laboratory personnel, while sprayed samples were produced by an external and certified shotcrete operator. During the first 24 hours, the specimens were protected using plastic sheeting, see Figure 2. Thereafter, the specimens were stored for the remaining 27 days before testing in a controlled environment in the laboratory.

3.2 *Evaluation of structural performance*

All the tested fibres could fulfil the structural requirements usually put on fibre-reinforced shotcrete, i.e. reach a target energy absorption or specific flexural residual strength (Sjölander et al., 2022a). Hence, all the tested fibres are suitable for rock support from a structural point of view. However, the fibres have different characteristics which must be acknowledged in the design. This is highlighted by the results presented in Figure 3. Here, the residual flexural strength at 2 mm deflection $f_{f1,re2}$ and the energy absorption at 40 mm vertical deflection E_{40} for cast beams and panels are shown. For the tested type of fibres, synthetic fibres are clearly less efficient at small displacements since the three tested dosages resulted in the lowest $f_{f1,re2}$ for all fibres. For large deformations, synthetic fibres are much more efficient and panels with a dosage of 6 and 9 kg/m^3 show the highest energy absorption of all tested fibre type and dosages. Steel fibres performs well at small and large displacement. While comparing steel and basalt fibres, the results indicate that steel performs better than basalt at large displacements. This can be seen by comparing the results for 14 kg/m^3 of basalt to 20 kg/m^3 of Dramix 4D. Both fibres show similar results for $f_{f1,re2}$ but the mean value for energy absorption at 40 mm, i.e. E_{40}, is significantly higher for Dramix 4D.

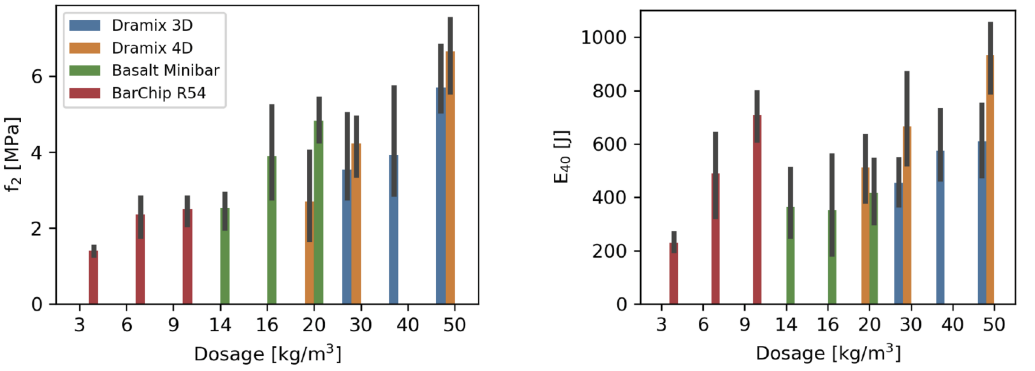


Figure 3. Flexural residual strength at 2 mm vertical deformation and energy absorption at 40 mm deflection for cast beams and panels with various fibre type and dosages. From (Sjölander et al., 2022a).

Most tests presented in the literature are based on cast specimens since these are easier and cheaper to produce in the laboratory. However, the results in Figure 4 indicate that there is a significant difference between cast and sprayed specimens. Here, the difference in $f_{f1,re2}$ with steel fibres and E_{40} with basalt fibres are shown to the left and right side, respectively. Thus, this effect must be considered while evaluating structural performance and it should be further investigated whether or not it is affected by the type of fibre. One of the reasons for the difference in structural performance could here be that fibres were lost during spraying due to rebound from the hard surface. Another reason could be that fibres became unfavorably aligned when hitting the hard surface under high pressure. A comparison between the fibre

orientation for the different fibre types will be performed using computed tomography (CT), see e.g (Ansell, 2023).

3.3 *Evaluations of environmental impact*

In the work by (Anand, 2021) and (Sjölander et al., 2022a), the environmental impact of various type of fibres was investigated. A case study representing the design of rock support for a Swedish tunnel was used to investigate the environmental impact caused by the fibres. Here, only the design phase of the tunnel was included, i.e. possible deterioration of fibres and the need for maintenance was not included. The same shotcrete mix and thickness was used, and the required dosage of fibres was determined based on different national design recommendations. For more details, see (Sjölander et al., 2022a). The environmental impact from the various fibres was based on available environmental product declarations (EPD) for each of the fibre

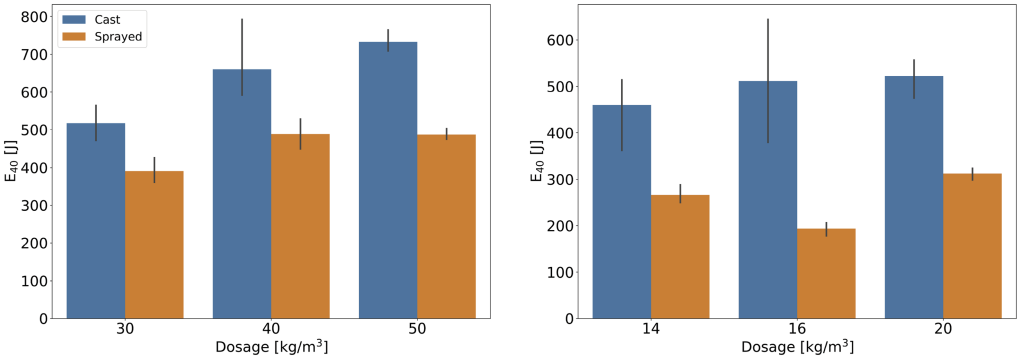


Figure 4. Energy absorption at 40 mm deflection for cast and sprayed panels with Dramix 3D steel fibres (left) and basalt Minibar fibres (right).

types used. Here, the equivalent emission of greenhouse gases from cradle to gate was used to compare the different fibres. This means that all emissions until 1 kg of fibres are ready to be shipped from the factory are included in the analysis. In Figure 5, a comparison between CO_2 emissions for the different fibre types is presented for two design cases. To the left, the lowest dosage of each fibre type that fulfilled a minimum $f_{f1,re2}$ of 3 MPa and the corresponding CO_2 emissions are shown. To the right, dosages required to fulfill a minimum E_{40} of 400 J (here

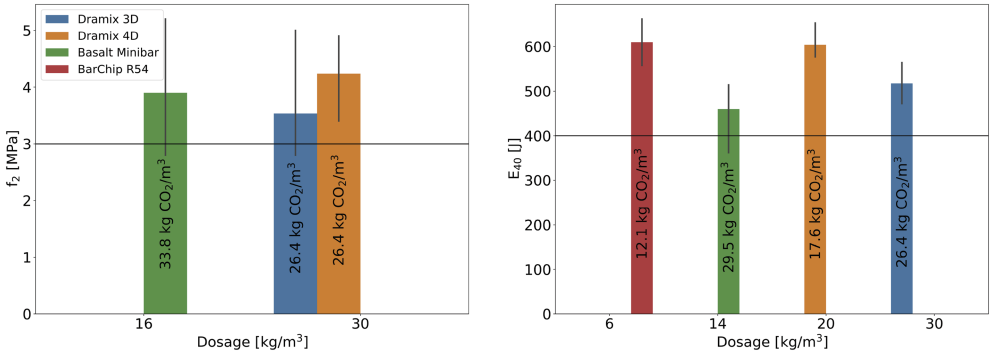


Figure 5. Emission of greenhouse gases for selected dosages of fibres that fulfil a flexural residual strength of 3 MPa at 2 mm vertical deflection (left) and an equivalent energy absorption of 1000 J at 40 mm deflection (right).

assumed equivalent to 1000 J for a rectangular slab used in the Q-system) is shown with the corresponding CO_2 emissions. First, it could be noted that the tested synthetic fibres did not fulfill a $f_{f1,re2}$ of 3 MPa and in this cases, steel fibres yield the lowest environmental impact, in terms of CO_2 emissions. For a design based on energy absorption, the synthetic fibres is clearly showing the greatest potential in lowering the CO_2 emissions. However, other aspects such as durability, effects due to long-term loading and how rebounded fibres containing synthetic material should be prevented to end up in the nature must also be investigated.

4 CONCLUSIONS AND FUTURE WORK

In this paper, results from structural testing of cast and sprayed fibre-reinforced shotcrete specimens have been presented. It was shown that all the tested fibre types were able to fulfil the structural requirements that are normally put on the rock support. However, different types of fibres are more or less suitable depending on expected behaviour of the rock mass. Moreover, it was shown that a large difference exists between cast and sprayed specimens. This could simply be due to the loss or unfavourable orientation of the fibres caused by the spraying process. This should be further investigated using computed tomography at a later stage.

The biggest challenge and obstacle to overcome before allowing fibres of alternative materials on the Swedish market is the durability. Steel suffers from corrosion which mainly is a problem in cracked sections and therefore, a problem that could be visually detected. Basalt fibres suffer from degradation in an alkaline environment, i.e. inside the shotcrete. However, the basalt is protected by a coating to prevent a contact with the concrete. But to ensure a technical life time of 120 years, the durability of the coating and how this is affected by potential cracks should be further investigated. A similar question arises for synthetic fibres, weather the material is affected by the alkaline environment during the life-time of the tunnel, which normally is 120 years. Also, if the presence of ground water and chlorides affect the durability of alternative fibre types must also be investigated.

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REFERENCES

Anand, S. (2021). *Evaluation of environmental and technical performance of alternate fibres for shotcrete in tunnels*. Master of Science Thesis. Stockholm, KTH Royal Institute of Technology.

Ansell, A. (1999). *Dynamically loaded rock reinforcement*. Doctoral Thesis. Stockholm, KTH Royal Institute of Technology.

Ansell, A. (2021). *Computed tomography as investigation method to describe the steel fibre distribution in sprayed and cast concrete for tunnels*. Technical Report. Stockholm, BeFo Rock Engineering Research Foundation.

Ansell, A. (2023). Tomography based method for investigation of steel fibre distribution in tunnel shotcrete, *In Proceedings of the World Tunnel Congress*. Athens, Greece.

ASTM. (2021). *C1550-19 Standard test method for flexural toughness of fibre-reinfroced concrete*. Technical standard. Pennsylvania, American Society for Testing and Materials.

Barrett, S. V. L. & McCreath, D. R. (1995). Shortcrete support design in blocky ground: Towards a deterministic approach, *Tunnelling and Underground Space Technology*. 10: pp 79–89.

Barros, J. A. O. & Cuncha, V. M. C. F. & Ribeiro, A. F. & Antunes, J. A. B. (2005). Post-cracking behaviour of steel fibre reinforced concrete, *Materials and Structures*. 38: pp 47–56.

Barton, N. & Lien, R. & Lunde, J (1974). Engineering classification of rock masses for the design of tunnel support, *Rock Mechanics*. 6: pp 189–236.

Bernard, E. S. (2002). Correlations in the behaviour of fibre reinforced shotcrete beam and panel specimens, *Materials and Structures*. 35: pp 156–164.

Bieniawski, Z.T. (1973). Engineering classification of jointed rock masses, *Civil Engineer in South Africa*. 15(12).
Brodd, E. and Östlund, L. (2022). *Environmental and technical evaluation of cement reduction and test methods for fibre reinforced shotcrete in tunnels*. Master of Science Thesis. Stockholm, KTH Royal Institute of Technology.
CEN. (2002). *Testing hardened concrete – Part 3: Compressive strength of test specimens*. Technical standard. Brussels, European Committee for Standardisation.
CEN. (2004). *Eurocode 7: Geotechnical design -Part 1 General rules*. Technical standard. Brussels, European Committee for Standardisation.
CEN. (2006). *EN 14488-5: Testing sprayed concrete - Part 3: Flexural strengths (first peak, ultimate and residual) of fibre reinforced beam specimens*. Technical standard. Brussels, European Committee for Standardisation.
CEN. (2006). *EN 14488-5: Testing sprayed concrete - Part 5: Determination of energy absorption capacity of fibre reinforced slab specimens*. Technical standard. Brussels, European Committee for Standardisation.
Holmgren, J. (1979). *Punch-loaded shotcrete linings on hard rock*. Technical Report. Stockholm, BeFo Rock Engineering Research Foundation.
Malmgren, L. (2005). *Interaction between shotcrete and rock*. Doctoral Thesis. Luleå, Luleå University of Technology.
Mohaghegh, A. (2018). *Structural performance of high-performance macro basalt fibre concrete: Flexural, shear, punching shear and fire spalling*. Doctoral Thesis. Stockholm, KTH Royal Institute of Technology.
Myren, S. & Hagelia, P. & Bjontegaard, O. (2018). The ban of polymer fibre in FRSC in Norwegian road tunnels. *In Proceedings of the 8th International Symposium on Sprayed Concrete*. Trondheim, Norway.
Nordström, E. (2005). *Durability of sprayed concrete*. Doctoral Thesis. Luleå, Luleå University of Technology.
Johansson, F. & Roslin, M. & Josefsson J. (2020). Evaluation of glass-fibre bolts as permanent rock support (in Swedish). Technical Report. Stockholm. KTH Royal Institute of Technology.
Rengarajan, M. (2020). Laboratory testing of shotcrete with fibres of steel, basalt or synthetic materials. Master of Science Thesis. Stockholm. KTH Royal Institute of Technology.
Sjölander, A. (2020). *Structural behaviour of shotcrete in hard rock tunnels*. Doctoral Thesis. Stockholm, KTH Royal Institute of Technology.
Sjölander, A. & Ansell, A. & Nordström, E. (2022). On the design of permanent rock support using fibre-reinforced shotcrete. *Submitted to: Fibers*.
Sjölander, A. & Ansell, A. & Nordström, E. (2022). Data from structural testing of sprayed and cast shotcrete reinforced with fibres of steel, basalt and synthetic material. *Mendeley Data* V1. doi: 10.17632/d7n5mvb2sg.